

# QEM-7: Energy-Field Coupling and Feedback in Mirror-Flip Arrays

QEM-7: Energy-Field Coupling and Feedback in Mirror-Flip Arrays

QEM-7量子纠缠力学第七卷

Author: Youfu Wang (王佑福)

ORCID: 0009-0005-5998-6024

Date: 2025-11-07

License: CC BY 4.0

Abstract (EN):

We extend the QEM-6 Mirror-Flip Unit (MFU) to device arrays and introduce an energy-field coupling model with readout-feedback control. Linearization around stationary domain-wall textures yields a Laplacian-spectrum-governed stability condition; we derive conservative bounds for feedback gain and delay, and generalize the flip-wave propagation limit to arrays. Three falsifiable protocols are provided: energy-flux pump-probe, feedback stability-island mapping, and array response spectroscopy. The results set timing budget, fan-out limits and safe operating windows for chip-oriented, controllable entanglement networks.

摘要 (ZH):

本文扩展了 QEM-6 镜像翻转单元 (MFU) 到器件阵列，并引入能量-场耦合模型，结合读出-反馈控制。围绕静态畴壁纹理的线性化分析给出了由拉普拉斯谱控制的稳定性条件；我们推导了反馈增益和延迟的保守性边界，并将翻转波传播极限推广到阵列。本文提供了三个可证伪的协议：能量-通量泵-探测、反馈稳定性-岛映射以及阵列响应光谱学。研究结果设定了时序预算、扇出限制以及芯片级可控纠缠网络的可靠操作窗口。

Series note: This work is part of the Quantum Entanglement Mechanics (QEM) series by Youfu Wang (2025).

# 1. Introduction

QEM-4: domain-wall stabilized textures; QEM-5: causal front bound; QEM-6: MFU minimal cell. QEM-7 upgrades MFU to arrays with energy-field coupling and feedback.

## 2. MFU Array and Energy Field

Variables per unit  $i$ :  $\phi_i$  ( $\Delta$ -phase),  $\psi_i$  (coherence),  $\epsilon_i$  (local energy),  $h_i(t)$  (drive/feedback).

Adjacency  $W$ , degree  $D$ , Laplacian  $L = D - W$ .

Local energy:  $\epsilon_i = 1/2 C (\dot{\phi}_i)^2 + 1/2 S (\nabla \phi_i)^2 + V(\phi_i) + \lambda \psi_i^2$ ; neighbor coupling  $K_{ij}(\phi_i - \phi_j)^2$ .

Energy flux:  $J_{ij} = -K_{ij} (\phi_i - \phi_j) \dot{\phi}_i$ .

### 3. Linearized Model and Laplacian Spectrum

Linearized dynamics near a stationary texture:  $\ddot{\phi} + \Gamma \dot{\phi} + \Omega_0^2 \phi + c_{\text{eff}}^2 L \phi = B h(t)$ .

Feedback closure:  $h(t) = g_f H \phi(t - \tau) + g_n W H \phi(t - \tau_n)$ .

Mode-wise polynomial:  $s_k^2 + \Gamma s_k + \Omega_0^2 + c_{\text{eff}}^2 \lambda_k - (g_f + g_n \lambda_k) H e^{-s_k \tau} = 0$ .

## 4. Feedback Stability Island and Design Rules

Conservative bounds (small-delay):  $g_{f^*} \lesssim (\Omega_0^2 + c_{\text{eff}}^2 \lambda_2)/|H|$ ,  $\tau < 0.2 \xi / c_{\text{eff}}$ .

Design:  $K_{\text{inter}}/J \leq 0.2$ ;  $\kappa/J \in [0.3, 0.5]$ ; keep feedback local; fanout  $\lesssim (\Omega_0^2 / ||W||)/g_n$ .

Timing:  $T_{\text{ro}} \ll \xi / c_{\text{eff}}$ ; feedback bandwidth covers  $[0, \Omega_0]$ .

## 5. Propagation Bound in Arrays

Array limit:  $v_{\max}^{\text{array}} \approx \min( v_{\max}^{\text{local}}, (a/\pi) \sqrt{\Omega_0^2 + c_{\text{eff}}^2 \lambda_{\max}} )$ .

$a$  is unit spacing; sets safe upper bound for front tracking.

## 6. Protocols and Metrics

P1 pump-probe energy flux; P2 stability-island mapping; P3 array spectroscopy.

Criteria:  $\pi$ -echo sustained; envelope  $\sim \text{sech}^2$ ;  $v_{\text{front}} \leq v_{\text{max}}^{\{\text{array}\}}$ .

## 7. Discussion and Outlook

QEM-7 bridges minimal MFUs to chip-level synthesis and prepares unified equations for QEM-8.

# References

Lieb & Robinson (1972); Cheneau et al. (2012); Calabrese & Cardy (2006); kink/sech<sup>2</sup> standard references.

Series note: This work is part of the Quantum Entanglement Mechanics (QEM) series by Youfu Wang (2025). | License: CC BY 4.0